Abnik Ahilasamy

AI Evaluation and Benchmarking

Email • Portfolio • GitHub • LinkedIn • X

Profile

I build and evaluate agentic AI systems, with a focus on whether models use tools reliably, preserve context across multi-step tasks, and ground answers in the information they retrieve. My work combines benchmark and task design, evaluation data, scoring, failure analysis, and practical system engineering.

Areas of focus

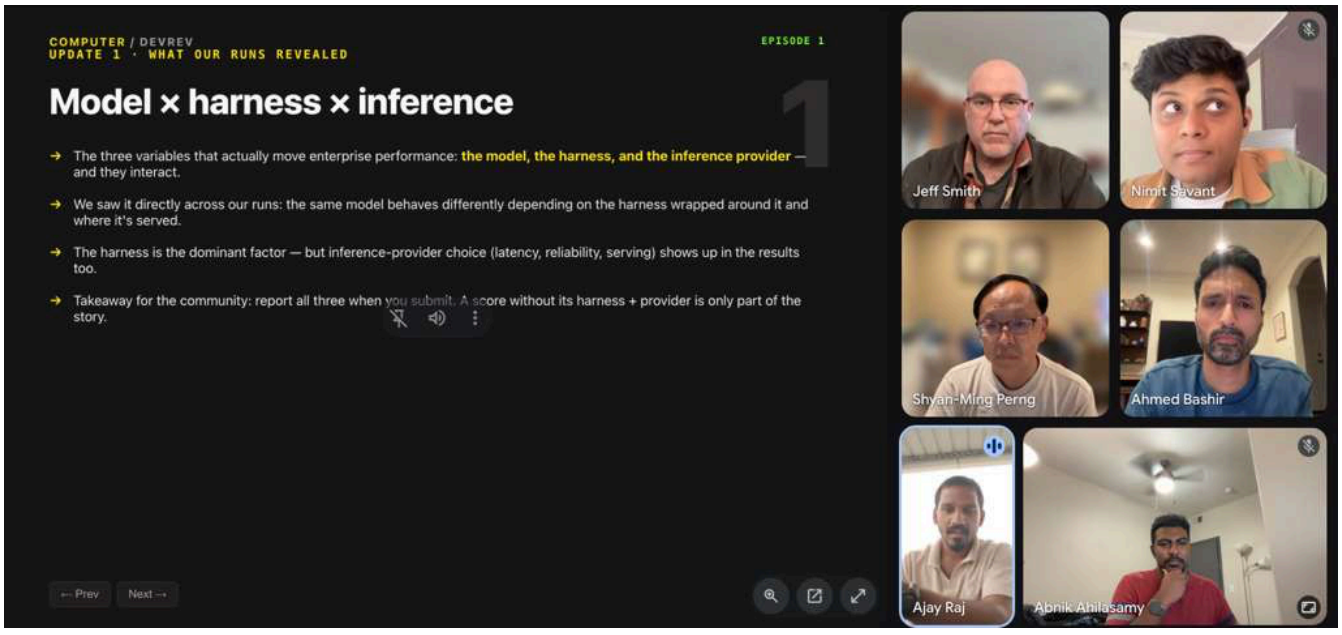
Agent benchmarks and longer-horizon tasks • Tool-use evaluation • Gold data and dataset quality • Deterministic and model-based scoring • Traces, reliability, and safety

Enterprise Bench with DevRev (unicorn startup)

I contributed task-design feedback and evaluation analysis to DevRev's open Enterprise-Bench, including ideas for multi-step and longer-horizon tasks. I reviewed L2 tasks and piloted five with GPT-5.6 Luna in one run; four passed. I examined a late-email failure and suggested virtual-clock checks, sandbox safeguards, and clearer run visualizations. I discussed this work with Jeff Smith, Ahmed DevRev's CTO, and members of the team during an official livestream.

https://devrev.ai/enterprise-bench-methodology

https://devrev.ai/blog/enterprise-bench



Discussing Enterprise-Bench with DevRev's team

Enterprise-Bench repository

Watch the DevRev livestream from 22:00 where I was introduced on their official YouTube page

Evaluation and Benchmark Work

Legal AI evaluation data pipeline at CIPS AI Labs

I built a reproducible dataset pipeline for legal-AI post-training evaluation across 873 Indian Supreme Court land-dispute cases. I organized the data into 12 evaluation views spanning eight judgment sections and four precedent-sentiment categories. The pipeline produced more than 4.19 million words across 790 content-bearing records, with 100% case-ID coverage and zero processing errors. It tracks 83 legitimately unsegmented cases separately and uses smoke tests, validation gates, resumable checkpoints, and recovery steps for long-running jobs.

SpotDiff Eval visual reasoning benchmark

I created an open development benchmark for models asked to identify visible differences between paired image panels. The current set contains 10 composite images and 61 annotated differences. A CLI accepts structured JSON predictions and deterministically matches them to gold labels, reporting precision, recall, coverage, attribute accuracy, missed items, extra predictions, and per-image results. I evaluated seven vision models end-to-end, smoke-tested eight more, and added 19 automated tests plus a perfect-baseline regression fixture. In one reported run, Gemini 3.1 Flash scored 32.8% overall, with 57.1% precision and 19.7% recall.

GitHub repository • Hugging Face development dataset

Awesome AI Evals resource collection

I am building an open, evolving index of practical resources on evaluation methods, metrics, datasets, benchmarks, agent evaluation, safety testing, and evaluation tools. The goal is to make it easier to find useful material and keep learning in one place.

GitHub repository

RunSentry agent observability and evaluation platform [Discontinued project]

I am developing an open platform that ingests standard OpenTelemetry traces over OTLP/gRPC or HTTP, removes sensitive content before durable storage, normalizes framework telemetry, reconstructs traces in a UI, and supports evidence-linked evaluations. The repository currently contains its first working vertical slice; online evaluation, clustering, diagnosis, and other later phases remain on the roadmap.

GitHub repository

Applied AI Systems and Demos

Aker RAG property-scoped AI assistant

I built a property-management assistant that combines deterministic SQL templates, a governed LLM-to-SQL fallback, and retrieval-augmented generation. Each request stays scoped to the active property across structured queries and vector retrieval. Generated SQL is checked before execution, and responses include citations. The service uses FastAPI, MySQL, Qdrant, and LangGraph.

GitHub repository • Architecture walkthrough

Freelance client project performance AI chat proof of concept

Through freelance work, I created a client-specific proof of concept for AI chat infrastructure. It includes a FastAPI orchestrator, an example MCP tool server, an OpenTelemetry collection path, UI shells, and Helm deployment scaffolding for local Kubernetes. The repository describes an early-stage architecture demo: agent services and the UI-to-orchestrator flow are still in progress.

GitHub demo repository

GitHub and Jira state sync

I built a webhook-driven service that links GitHub pull requests with Jira issues and Slack alerts. It posts a Jira comment when a pull request opens, checks GitHub and Jira state again after a delay, and uses YAML policy rules to decide when to send a mismatch alert to Slack.

GitHub repository • Watch the workflow demo

AffectKit safety-aware agent middleware

I built an open Python middleware package that gives agents a configurable, simulated affective state across turns. It includes profile-driven state updates, mood decay, and adapters for plain Python, LangChain, and LangGraph. Safety rules shape the behavior and keep simulated affect separate from claims of sentience.

GitHub repository

Experience and Background

Post-training researcher at CIPS AI Labs

June 2026 to present • I work on repeatable legal-AI data and evaluation pipelines for post-training research.

LLM engineer at Larsen and Toubro

I built and benchmarked an agentic RAG service using LangChain and LlamaIndex. On a fixed 100-query evaluation set, retrieval and inference improvements reduced p95 latency by 40% and hallucination rate by 35%. I added faithfulness and groundedness metrics, safety checks, fallbacks, and OpenTelemetry tracing; unsafe outputs fell by 50% across 50 safety cases.

Earlier machine learning and analytics internships

At EXCELERATE, I standardized reporting and validation workflows, improving decision efficiency by 30% and reducing analysis turnaround by 25%. At LetsGrowMore, I benchmarked more than five ML projects and reported 15–20% improved sequence prediction and over 90% CNN validation accuracy.

Education and research

Arizona State University • M.S. in Data Science • May 2026 • GPA 3.97/4.0

SRM Institute of Science and Technology • B.Tech. in Artificial Intelligence and Machine Learning • June 2024 • GPA 9.3/10 (3.8/4.0 equivalent) • Top 5% of cohort

Publication: “Brain Tumor Classification with Transfer Learning” using ResNet-50, IEEE 2023.

Writing and interests

I write about practical AI systems, RAG, agent reliability, evaluation, and deployment.

On X, I share what I’m learning, ideas I’m exploring, and progress on things I build. I am very active here.

Blog • X • Portfolio

I enjoy turning open-ended problems into testable questions, tracing failures to their causes, and building small tools that make the next experiment easier. Outside of work, I like swimming, watching films, and spending time with friends.

Tools and methods

Evaluation methods: benchmark and task design, dataset curation, gold-label alignment, deterministic scoring, model-based grading, failure analysis, regression testing, and safety evaluation.